Travel Constraint Extraction

Fine-tuning a small Transformer to understand changes to a London itinerary

A 400-example dataset for structured extraction

Task and output contract

“Keep the gallery. Be back by 19:00.
I would prefer less walking.”

Meaning	Structured output	Priority
Keep the gallery	Include the resolved place	Hard
Return by 19:00	hotel_return <= 19:00	Hard
Less walking	walking_amount: minimize	Soft

Missing time or walking scope requires clarification

Data and LoRA training

400 examples

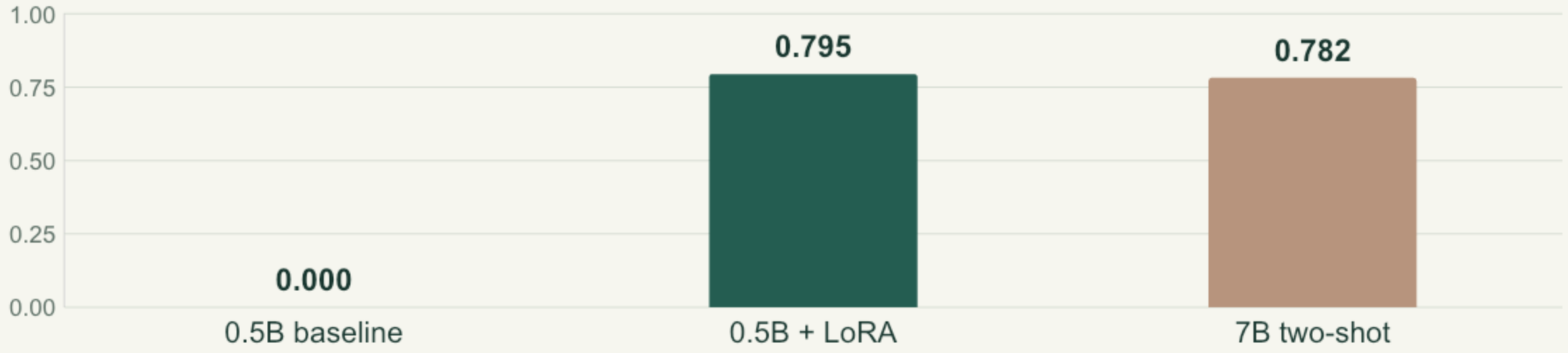
280 train / 40 validation / 80 test

100 scenarios × 4 paraphrases, grouped before splitting

Approach	Model and adaptation
Small baseline	Qwen2.5-0.5B, zero-shot
Fine-tuned small model	Same 0.5B + LoRA, rank 8, 840 steps
Larger comparison	Qwen2.5-7B, 4-bit, two training examples

AI-reviewed synthetic labels: zero human-verified records

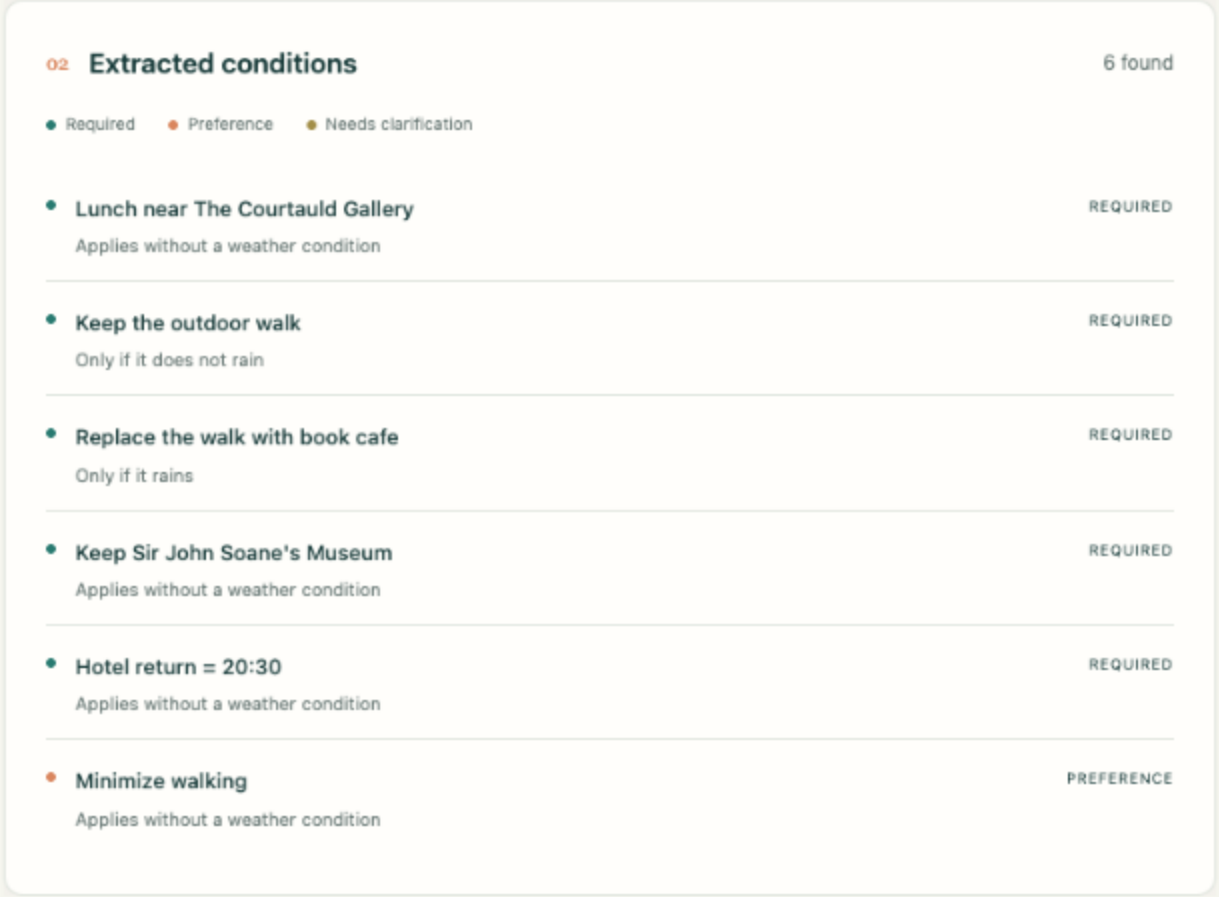
Held-out extraction results



Approach	Exact records	Valid JSON
0.5B baseline	0 / 80	0.0%
0.5B + LoRA	57 / 80	97.5%
7B two-shot	51 / 80	93.8%

80 inputs in 20 groups. Strict JSON without repair. Different quantization and prompting.

Conditional London demonstration



02 Extracted conditions — 6 found

Required · Preference · Needs clarification

- **Lunch near The Courtauld Gallery** — REQUIRED
  Applies without a weather condition
- **Keep the outdoor walk** — REQUIRED
  Only if it does not rain
- **Replace the walk with book cafe** — REQUIRED
  Only if it rains
- **Keep Sir John Soane's Museum** — REQUIRED
  Applies without a weather condition
- **Hotel return = 20:30** — REQUIRED
  Applies without a weather condition
- **Minimize walking** — PREFERENCE
  Applies without a weather condition

Forecast or simulation

Rain
Replace the outdoor walk
with a book-cafe category

Dry
Keep the planned walk

Unknown
Keep both branches pending

Actual saved validation output. Live mode uses the published LoRA adapter.

Limitations and next steps

0.795 F1 after LoRA fine-tuning

Measured on a controlled synthetic test set

Remaining model errors	Walking scope, time operators, and missed conditions
Next evaluation	Human-annotated requests across more scenarios
Application boundary	Illustrative edits with unverified routes and availability

github.com/hjarchivenote/travel-constraint-extraction